I The scalar case

$$J(x) = ax^2 + bx + c$$

$$= a\left(x^2 + \frac{b}{a}x + c\right)$$

Take the coefficient of x and divide by 2

$$= a\left(x + \frac{b}{2a}\right)^2 - \frac{b^2}{4a} + c$$

$$x = -\frac{b}{2a} \Rightarrow J_{min} = c - \frac{b^2}{4a}$$

A) Example

Demonstrate the use of completing the square for the estimation of a Gaussian random variable observed in Gaussian noise.

Suppose

Normal distribution, distribution

$$x \sim \mathcal{N}(\mu_x, \sigma_x^2) \quad \text{and} \quad N \sim \mathcal{N}(0, \sigma_n^2)$$

mean of 0

where

x is regularded as signal

N is the noise

The signal in noise is

$$Y = x + N$$

Probability density of x given y has occurred

Given a measure $Y = y$, we want to find $f(x|y)$
Using Bayes Theorem

$$f(x|y) = \frac{f(y|x)\,f(x)}{\int_{-\infty}^{\infty} f(x|y)\,f(x)\,dx}$$

For some signal $X = x$ we can write

$$Y = x + N$$

therefore $N = Y - x$. That is $f(x|y) = f_n(y-x)$

Density of white noise

The previous ~~statement~~ can be written

$$f(x|y) = \frac{f_n(y-x)f(x)}{\int_{-\infty}^{\infty} f(x|y)f(x)dx}$$

$$= \frac{1}{C}\frac{1}{\sigma_n\sqrt{2\pi}}e^{-(y-x)^2/2\sigma_n^2}\frac{1}{\sigma_x\sqrt{2\pi}}e^{-(x-\mu_x)^2/2\sigma_x^2}$$

The exponents can be written as

$$E = x^2\left(-\frac{1}{2\sigma_n^2} - \frac{1}{2\sigma_x^2}\right) + x\left(\frac{y}{\sigma_n^2} + \frac{\mu_x}{\sigma_x^2}\right) + C_1$$

By completing the square

$$E = -\frac{\sigma_x^2+\sigma_n^2}{2\sigma_n^2\sigma_x^2}\left(x - \frac{y\sigma_x^2+\mu_x\sigma_n^2}{\sigma_x^2+\sigma_n^2}\right)^2 + C_2$$

$$\Rightarrow \quad f(x|y) = \frac{e^{C_2}}{2\pi C\sigma_n\sigma_x}e^{E}$$

Which has the form of a Gaussian density. Because of this the exponential must be such that this integrates to 1

$$\Rightarrow \quad \mu_{x|y} = \frac{\sigma_x^2}{\sigma_x^2+\sigma_n^2}y + \frac{\sigma_n^2}{\sigma_x^2+\sigma_n^2}\mu_x$$

$$\sigma_{x|y}^2 = \frac{\sigma_n^2\sigma_x^2}{\sigma_x^2+\sigma_n^2}$$

Lets interpret the results.

If $\sigma_n^2 >> \sigma_x^2$ then an observation Y does not tell us much about x because the noise N is too strong.

If $\sigma_n^2 << \sigma_x^2$ then an observation Y is about the same as the observation X.

II The matrix case

$$x^T A x + x^T y + c$$

and A is symmetric

$$(x+z)^T A (x+z) + d \quad \text{where} \quad z = \frac{1}{2} A^{-1} y$$

$$d = c - \frac{1}{2} y^T z$$